



JERUSALEM COLLEGE OF ENGINEERING
(An Autonomous Institution,
Affiliated to Anna University, Chennai)

QUESTION BANK

JAL2521 TIME SERIES ANALYSIS AND FORECASTING

Regulation : 2023

Branch : B.E. CSE (AI & ML)

Year / Semester : III / V



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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
QUESTION BANK

Regulation	:	2023
Course Name	:	TIME SERIES ANALYSIS AND FORECASTING
Course Code	:	JAL2521
Year / Semester / Branch	:	III / V / AI & ML
Unit No.	:	I
Unit Title	:	INTRODUCTION TO TIME SERIES AND FORECASTING
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART – A

S.No.	Question	Marks	BTL	CO
1.	What is time series analysis and forecasting? - Time series analysis and forecasting is the process of examining sequential data points collected or recorded over time to identify patterns, trends, and seasonal variations and using historical data and identified patterns to predict future values or behavior of the series. - Businesses and researchers can make data-driven decisions, enhance operational efficiency, and respond proactively to future challenges using time series analysis and forecasting.	2	K2	CO1
2.	List out the various domains in which time series analysis and forecasting are applied. - Finance and economics - Retail and supply chain - Technology - Healthcare - Energy and utilities - Manufacturing and industry - Transportation - Social media and marketing - Weather and climate monitoring	2	K2	CO1
3.	Write about few applications of time series analysis and forecasting. Stock Market Prediction Forecasting stock prices and market indices based on historical trends. Demand Forecasting Estimating future demand for products to optimize inventory levels and avoid overstocking or stockouts. Sales Forecasting Predicting revenue and sales patterns for better marketing strategies and resource allocation.	2	K2	CO1



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	• Patient Monitoring Analyzing time series data from wearable devices or medical equipment to detect health anomalies. • Load Forecasting Predicting electricity demand to optimize power generation and distribution. • Network Traffic Analysis Monitoring and forecasting data traffic for efficient resource allocation in IT infrastructure. • Fraud Detection Identifying unusual patterns in transactional data to prevent cyber fraud. • Sentiment Analysis Tracking sentiment trends over time to assess brand perception.												
4.	Mention the purpose of time series analysis. • To understand and interpret the underlying forces that produce the observed state of a system or process over time. • To forecast the future state of the system or process in terms of observable characteristics.	2	K2	CO1									
5.	Differentiate the different types of data. <table><tr><th><i>Cross-sectional Data</i></th><th><i>Time series Data</i></th><th><i>Panel Data</i></th></tr><tr><td>Data obtained by taking observations from multiple individuals at the same point or different points in time.</td><td>Data obtained from an individual entity, taken at multiple points in time.</td><td>Data obtained from multiple entities over multiple points in time.</td></tr><tr><td>Eg. Yearly military expenditure for different countries.</td><td>Eg. Yearly military expenditure for India.</td><td>Eg. Military expenditure considered for 4 countries over the same period of 1960-2010</td></tr></table>	<i>Cross-sectional Data</i>	<i>Time series Data</i>	<i>Panel Data</i>	Data obtained by taking observations from multiple individuals at the same point or different points in time.	Data obtained from an individual entity, taken at multiple points in time.	Data obtained from multiple entities over multiple points in time.	Eg. Yearly military expenditure for different countries.	Eg. Yearly military expenditure for India.	Eg. Military expenditure considered for 4 countries over the same period of 1960-2010	2	K1	CO1
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Eg. Yearly military expenditure for different countries.	Eg. Yearly military expenditure for India.	Eg. Military expenditure considered for 4 countries over the same period of 1960-2010											
6.	What is meant by internal structures of time series? Internal structures of time series are the special characteristics of time series data that requires its special mathematical treatment. They are: • General trend • Seasonality • Cyclical movements • Unexpected variations	2	K2	CO1									
7.	What does run sequence plot indicate? Run sequence plot indicates the following properties of the time series: • Movements in mean of the series • Shifts in variance • Presence of outliers	2	K2	CO1									



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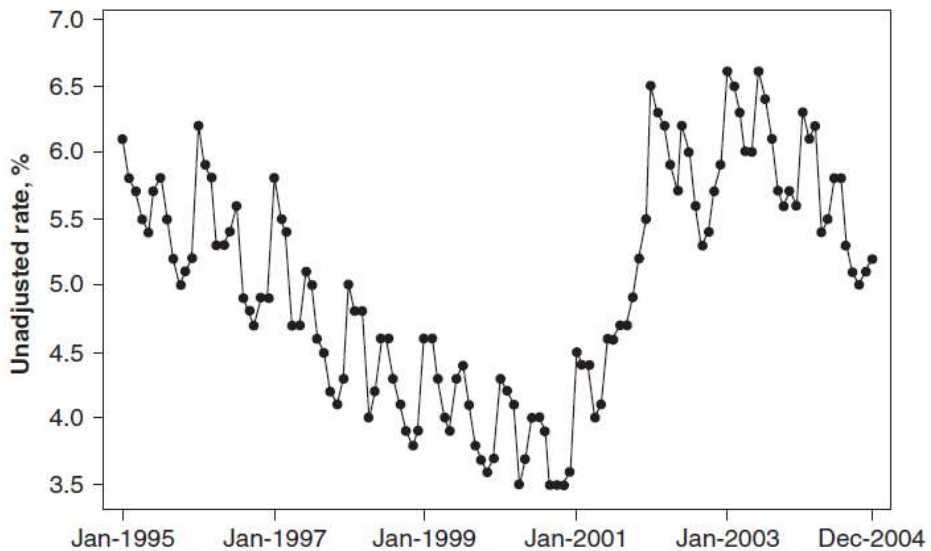
8.	Differentiate seasonality and cyclical changes.		2	K2	CO1
	Seasonality	Cyclical changes			
	Reveals the repetitive and period variations in a time series, e.g., daily, monthly, yearly.	Reveals irregular patterns that occur over longer, non-fixed durations linked to economic or business cycles.			
	Eg. observations of CO ₂ emissions, monthly sales volume of seasonal goods or clothing, ice cream sales in summer.	Eg. Economic growth and recession cycles, industry-specific booms.			
	Determined through EDA and run sequence plot, seasonal sub series plot and multiple box plots	Identified through a long run plot of the time series data.			
	Highly predictable due to fixed periodicity.	Less predictable due to irregular timing and duration.			
9.	What are the models that are often used as building blocks of time series analysis? • <i>Zero mean models</i> - models where the time series fluctuates around a constant mean and constant variance values. • <i>Random walk</i> - models where the time series fluctuates around a zero mean and constant variance values • <i>Trend models</i> – models that capture the long run trend in the time series that can be fitted as linear regression of the time index • <i>Seasonality models</i> – models that reveal the periodic and repetitive fluctuations in a time series and hence are modeled as sum of weighted sum of sine waves of known periodicity.		2	K1	CO1
10.	Give the generic approach of a time series analysis. Step 1: Visualize the data at different granularities of the time index to reveal long run trends and seasonal fluctuations. Step 2: Fit trend line capture long run trends and plot the residuals to check for seasonality or irreducible error. Step 3: Fit a harmonic regression model to capture seasonality. Step 4: Plot the residuals left by the seasonality model to check for irreducible error.		2	K1	CO1
11.	Define autocorrelation. Autocorrelation, also known as serial correlation, is a statistical measure of the relationship between a time series and a lagged version of itself. • It quantifies how the current value in a series is correlated with its past values. • Autocorrelation values range from −1 to +1.		2	K1	CO1



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12.	How does autocorrelation differ from partial autocorrelation? Autocorrelation measures the correlation between a time series and its lagged values, considering all intermediate lags. Partial autocorrelation measures the correlation between a time series and its lagged values while controlling for the influence of intermediate lags.	2	K2	CO1
13.	What are autoregressive models? AR models are statistical models that predict future values based on past values in the series. They assume that past observations have a linear relationship with current values.	2	K2	CO1
14.	Differentiate qualitative and quantitative forecasting techniques. Qualitative forecasting is based on subjective judgment, intuition, and expertise. Often used when data is scarce or unavailable. Quantitative forecasting is based on historical data and statistical analysis, using mathematical models to predict future values.	2	K2	CO1
15.	Draw an example time series plot.  Monthly unemployment rate—full-time labor force, unadjusted. Exhibits a mixture of patterns	2	K2	CO1
16.	What are the three most widely used quantitative forecasting models? • Regression models • Smoothing models • Time series models	2	K1	CO1
17.	List out the activities in the forecasting process. The activities in the forecasting process are: 1. Problem definition 5. Model validation 2. Data collection 6. Forecasting model deployment 3. Data analysis 7. Monitoring forecasting model performance	2	K2	CO1



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	4. Model selection and fitting															
18.	What is the role of data warehouse in time series analysis and forecasting? A data warehouse plays a critical role in ensuring the efficiency and accuracy of data analysis and model building. It helps through: - Centralized data storage - Ensuring data quality and consistency - Historical data access - Efficient querying and analysis - Data aggregation - Scalability and flexibility - Scalability and - Data security and compliance	2	K2	CO1												
19.	What are the different types of data imputation? <table><tr><th>TYPE OF IMPUTATION</th><th>DESCRIPTION</th></tr><tr><td>Mean value imputation</td><td>An easy method that replaces a missing value with the sample average calculated from the non-missing observations.</td></tr><tr><td>Stochastic mean value imputation</td><td>A random variable is added to the mean value to capture some of the noise or variability in the data.</td></tr><tr><td>Regression imputation</td><td>The imputed value is computed from a model used to predict the missing value. The prediction model does not have to be a linear regression model.</td></tr><tr><td>Hot deck imputation</td><td>The missing values are imputed by using values from similar complete observations.</td></tr><tr><td>Cold deck imputation</td><td>Uses information from a deck of cards not currently in use.</td></tr></table>	TYPE OF IMPUTATION	DESCRIPTION	Mean value imputation	An easy method that replaces a missing value with the sample average calculated from the non-missing observations.	Stochastic mean value imputation	A random variable is added to the mean value to capture some of the noise or variability in the data.	Regression imputation	The imputed value is computed from a model used to predict the missing value. The prediction model does not have to be a linear regression model.	Hot deck imputation	The missing values are imputed by using values from similar complete observations.	Cold deck imputation	Uses information from a deck of cards not currently in use.	2		CO1
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20.	Define data cleaning. Data cleaning is the process of examining data to detect potential errors, missing data, outliers or unusual values, or other inconsistencies and then correcting the errors or problems that are found.	2	K2	CO1												
21.	What is data imputation? What is its purpose? Data imputation is the process of correcting missing data or replacing outliers with an estimation process. Imputation replaces missing or erroneous values with a “likely” value based on other available information. It enables the analysis to work with statistical techniques which are designed to handle the complete data sets.	2	K2	CO1												



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PART – B

S. No.	Question	Marks	BTL	CO
1.	What are the different types of data used in analytics? Explain their purpose with suitable examples.	16	K2	CO1
2.	Discuss the internal structure of a time series. Explain trend, seasonality, cyclical, and irregular components with suitable illustrations	16	K2	CO1
3.	Write in detail about the general models that are often used as building blocks of time series analysis.	16	K2	CO1
4.	Compare and contrast the following with examples: (i) Autocorrelation and partial autocorrelation. (ii) Qualitative and quantitative forecasting methods	16	K2	CO1
5.	Write a short note on the nature and uses of forecasting.	16	K1	CO1
6.	List down and explain each of the activities in the forecasting process.	16	K2	CO1
7.	Explain about the role of data warehouses, data cleaning and imputation in time series analysis.	16	K1	CO1



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Unit No.	:	II
Unit Title	:	STATISTICS BACKGROUND FOR FORECASTING
No. of Hours	:	9
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PART – A

S.No.	Question	Marks	BTL	CO												
1.	Differentiate between fitted value and forecast value.	2	K2	CO2												
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Prediction for future data beyond the observed range.	Estimation for past/observed data.															
2.	What is lead – 1 forecast and lead – 1 forecast error? • Lead-1 forecast is the forecast of the observation in period t, y_t, made one period prior, $\hat{y}_t(t - 1)$.	2	K2	CO2												



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	The forecast error that results from a forecast of y_t that was made at time period $t - \tau$ is the lead - τ forecast error; if $\tau=1$, it is lead-1 forecast error. $e_t(1) = y_t - \hat{y}_t(t-1)$			
3.	What are forecast and residual errors? Forecast error - Difference between actual future values and forecasts. $e_t(\tau) = y_t - \hat{y}_t(t-\tau)$, where τ is the lag. Residual error - Difference between actual and fitted values. $e_t = y_t - \hat{y}_t$	2	K2	CO2
4.	What is a time series plot? When is scatter plot used? - Time series plot is just a graph of y_t versus the time period, t, for $t = 1, 2, \dots, T$. Features such as trend and seasonality are usually easy to see from the time series plot. - When there are two or more variables of interest, scatter plots can be useful in displaying the relationship between the variables.	2	K1	CO2
5.	What are data smoothers? Mention the different types of data smoothers. Data smoothers (or) data smoothing techniques help reduce noise in time series data and reveal underlying patterns. Different types of data smoothers are 1. <i>Simple moving average or linear data smoother</i> - calculates the average of a fixed number of past observations. $M_T = \frac{y_T + y_{T-1} + \dots + y_{T-N+1}}{N} = \frac{1}{N} \sum_{t=T-N+1}^T y_t$ 2. <i>Odd-span moving medians or running medians</i> - takes the median of a set number of past observations. The moving median for a span of $N=2u+1$, $m_t^{[N]} = \text{med}(y_{t-u}, \dots, y_t, \dots, y_{t+u})$	2	K2	CO2
6.	What is a stationary time series? A time series is said to be strictly stationary if its properties are not affected by a change in the time origin. If joint probability distribution of the observations $y_t, y_{t+1}, \dots, y_{t+n}$ is the same as the joint probability distribution of the observations $y_{t+k}, y_{t+k+1}, \dots, y_{t+k+n}$ then the time series is a stationary time series.	2	K2	CO2
7.	What are autocovariance and autocovariance functions? The covariance between y_t and its value at another time period, say, y_{t+k} is called the autocovariance at lag k . it is defined by	2	K1	CO2



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	$\gamma_k = \text{Cov}(y_t, y_{t+k}) = E[(y_t - \mu)(y_{t+k} - \mu)]$ The collection of the values of γ_k, for $k = 0, 1, 2, \dots$ is called the autocovariance function.			
8.	What are autocorrelation and autocorrelation functions? Autocorrelation coefficient at lag k for a stationary time series measures the correlation between observations separated by lag k. $\rho_k = \frac{E[(y_t - \mu)(y_{t+k} - \mu)]}{\sqrt{E[(y_t - \mu)^2]E[(y_{t+k} - \mu)^2]}} = \frac{\text{Cov}(y_t, y_{t+k})}{\text{Var}(y_t)} = \frac{\gamma_k}{\gamma_0}$ The collection of the values of ρ_k, for $k = 0, 1, 2, \dots$ is called the autocorrelation function.	2	K1	CO2
9.	What are the techniques for determining if a time series is nonstationary? - Plotting a reasonable long series of the data to see if it drifts or wanders away from its mean for long periods of time, - Computing the sample ACF.	2	K2	CO2
10.	What is a variogram? Give an example. The variogram G_k is a diagnostic tool that measures variances of the differences between observations that are k lags apart ($k=0,1,2,\dots$), relative to the variance of the differences that are at lag 1 ($k=1$). Mathematically, $G_k = \frac{\text{Var}(y_{t+k} - y_t)}{\text{Var}(y_{t+1} - y_t)} \quad k = 1, 2, \dots$ For a stationary time series, where ρ_k is the autocorrelation coefficient at lag k. $G_k = \frac{1 - \rho_k}{1 - \rho_1}$ Example:	2	K1	CO2
11.	What is data transformation in time series analysis and forecasting? Data transformation is the process of applying mathematical operations to the data to make it more suitable for modeling and forecasting. It is useful in statistical work for stabilizing the variance of the data.	2	K1	CO2



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12.	What is the power family of transformations? What are the different types of data transformations? Power family of transformations is used to deal with nonconstant variance. It is given by: $y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda y^{\lambda-1}}, & \lambda \neq 0 \\ \hat{y} \ln y, & \lambda = 0 \end{cases}$ The different types of data transformations is based on the value of λ: If $\lambda = 0.5$, it is a square root transformation. If $\lambda = 0$, it is the log transformation. If $\lambda = -0.5$, it is the reciprocal square root transformation. If $\lambda = -1$, it is inverse transformation.	2	K2	CO2
13.	What are trend adjustments and seasonal adjustments? Trend adjustment involves removing or modifying the long-term pattern in a dataset to better understand fluctuations due to seasonality or other factors. Seasonal adjustment removes periodic patterns (e.g., monthly or quarterly variations) to reveal the underlying trend and irregular fluctuations.	2	K2	CO2
14.	Write about difference and backshift operators. They are applied to remove trend. - Difference operator ∇ is applied for differencing the data to the original time series to obtain a new time series: $x_t = y_t - y_{t-1} = \nabla y_t$. - Backshift operator B is applied for differencing the data: $By_t = y_{t-1}$. So new time series, $x_t = (1 - B)y_t = \nabla y_t = y_t - y_{t-1}$ with $\nabla = (1 - B)$.	2	K2	CO2
15.	List down the steps in modeling and forecasting a time series. The basic steps in modeling and forecasting a time series are as follows: - Plot the time series and determine its basic features - Eliminate any trend or seasonal components - Develop a forecasting model for the residuals - Validate the performance of the model - Find the differences between the original time series the values that would be forecast by the model - Reverse the transformation if required. - Construct prediction intervals for the residuals, if required.	2	K2	CO2



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	8. Develop and implement a procedure for monitoring the forecast			
16.	What are the standard measures of forecast accuracy? 1. Average error or mean error 2. Mean absolute deviation (or mean absolute error) 3. Mean squared error	2	K1	CO2
17.	What is parsimonious approach? A parsimonious approach in time series analysis builds the simplest possible model with the fewest parameters that still provides an accurate forecast or fit.	2	K1	CO2
18.	What is meant by data splitting? Data splitting is a process of dividing the time series data into two segments - one for model fitting and the other for performance testing. It is also called as cross-validation. Good rule of thumb is to have at least 20 or 25 observations in the performance testing data set.	2	K1	CO2
19.	Mention few criteria for evaluating the fit of the model to historical data. • Mean squared error of the residuals • R-squared statistic • Adjusted R-squared statistic • Akaike Information Criterion (AIC) • Schwarz Bayesian Information Criterion (SBIC)	2	K2	CO2
20.	Why is performance monitoring important? Regardless of how well the model works initially, over time it is likely that its performance will deteriorate. The underlying pattern of the time series may change, either because the internal inertial forces that drive the process may evolve through time, or because of external events such as new customers entering the market. Hence performance monitoring is important.	2	K2	CO2
21.	What is Shewhart control chart? Shewhart control chart is the simplest way to monitor forecasting model performance. It is applied to the forecast errors. It is a plot of the forecast errors versus time containing a center line that represents the average (or the target value) of the forecast errors and a set of control limits that are designed to provide an indication that the forecasting model performance has changed.	2	K2	CO2



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PART – B

S. No	Question	Marks	BTL	CO
1.	Discuss about the role of graphical displays in developing a forecasting model and the use of time series plots and plotting smoothed data.	16	K2	CO2
2.	Discuss the numerical characteristics of time-series data. Explain how measures such as mean, variance, autocovariance, and autocorrelation describe the internal structure of a time series.	16	K2	CO2
3.	What is the purpose of data transformations in time series data? Describe the different types of data transformations.	16	K2	CO2
4.	Discuss in brief about trend and seasonal adjustments.	16	K1	CO2
5.	Describe the basic steps in modeling and forecasting a time series.	16	K2	CO2
6.	(i) Explain how the performance of a forecasting technique is evaluated for a particular time series or application. (ii) Explain the role of accuracy measures such as Mean Absolute Error, Mean Squared Error, and Mean Absolute Percentage Error	16	K2	CO2
7.	Explain in brief about the different ways of selecting an appropriate forecasting model.	16	K2	CO2
8.	Give the importance of monitoring the performance of a forecasting model.	16	K2	CO2



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UNIT III

REGRESSION ANALYSIS AND FORECASTING

Course Objective – To provide knowledge on the application of regression models for forecasting.

Course Outcome – Understand the fundamental principles of linear regression and its importance in modeling relationships between variables.

Bloom's Taxonomy Level – K1, K2

PART – A

S.No.	Question	Marks	BTL	CO
1.	Give the regression model that uses multiple predictors. Regression models that include more than one predictor or regressor variable is called as a the multiple linear regression model. If there are k predictors, the multiple linear regression model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \varepsilon$	2	K2	CO3
2.	Write the regression model for cross-section data and time series data. The regression model for cross-section data is written as $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n$ The regression model for time series data is written as $y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \dots + \beta_k x_{tk} + \varepsilon_t, \quad t = 1, 2, \dots, T$	2	K1	CO3
3.	What is least squares method?	2	K1	CO3



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	Least squares method is a technique to estimate the unknown model parameters $\beta_0, \beta_1, \dots, \beta_k$ (the β 's) in a linear regression model written as $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \varepsilon$			
4.	What is the statistical inference made in linear regression? The procedure for testing the significance of regression obtaining the confidence intervals requires to assume that the model errors are normally and independently distributed with mean zero and variance σ^2 , abbreviated NID(0, σ^2).	2	K2	CO3
5.	What is the test for significance of regression? Mention the hypotheses for the same. The test for significance of regression is a test to determine whether there is a linear relationship between the response variable y and a subset of the predictor or regressor variables $x_1, x_2, \dots, x_k$. The appropriate hypotheses are $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ $H_1 : \text{at least one } \beta_j \neq 0$	2	K2	CO3
6.	Differentiate prediction interval and confidence interval. - The confidence interval (CI) is an interval estimate on the mean of the response distribution at a specific point. - The prediction interval (PI) is an interval estimate on a single future observation from the response distribution at that point. - PI should be longer than the corresponding CI at the same point.	2	K2	CO3
7.	What are residual plots? Residual plots are the primary approach to model adequacy checking. They are used to check the adequacy of the normality assumption on the model errors by constructing a normal probability plot of the residuals.	2	K2	CO3
8.	What are standardized residuals? Standardized residuals are obtained from standardizing process that scales the residuals by dividing them by their approximate average standard deviation to. The standardized residuals have mean zero and approximately unit variance; consequently, they are useful in looking for outliers.	2	K2	CO3
9.	What are influential points? In building regression models, a small subset of the data can be found to exert a disproportionate influence on the fitted model. Estimates of the model parameters or predictions may depend more on the influential subset than on the majority of the data. They are	2	K2	CO3



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	called as influential points. They must be located and their impact on the model must be assessed.			
10.	Write a short note on the different methods used for variable selection in regression. <u>Stepwise type variable selection methods</u> → build a regression model by either adding or removing a predictor variable to the basic model at each step. <u>Forward selection</u> → the procedure begins with a model containing none of the candidate predictor variables and sequentially inserts variables into the model one at-a-time until a final equation is produced. <u>Backward elimination</u> → the procedure begins with all of the candidate predictor variables in the equation, and then variables are removed one-at-a-time to produce a final equation. <u>All possible regressions</u> → with K candidate predictor variables, the procedure examines all 2^K possible regression equations to identify the ones with potential to be a useful model.	2	K2	CO3
11.	What is weighted least squares (WLS) method? It is a method used to deal with nonconstant error variance is to fit the model. In this method of estimation the deviation between the observed and expected values of y_i is multiplied by a weight w_i that is inversely proportional to the variance of y_i. the WLS function is $L = \sum_{i=1}^n w_i (y_i - \beta_0 - \beta_1 x_i)^2$	2	K2	CO3
12.	What is Discounted Least Squares (DLS) method? The Discounted Least Squares (DLS) method is a variation of the ordinary least squares (OLS) technique used primarily in time series forecasting and adaptive filtering, where more recent observations are given greater weight than older ones.	2	K2	CO3
13.	Mention the effects of autocorrelation on OLS regression procedure. - Unbiased coefficients - Inefficient estimates - Underestimated or overestimated standard errors.	2	K1	CO3
14.	Write about Durbin Watson test. Durbin and Watson test is a statistical test to detect the presence of autocorrelation. It is based on the assumption that the errors in the	2	K1	CO3



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	regression model are generated by a first-order autoregressive process observed at equally spaced time periods.			
15.	State Cochrane–Orcutt method. The Cochrane–Orcutt method is a technique used to correct for autocorrelation in the residuals of a linear regression model. It transforms the model so that the autocorrelation is removed, and then re-estimates the regression using OLS on the transformed data.	2	K1	CO3

PART – B

S. No	Question	Marks	BTL	CO
1.	Describe how the least squares estimation procedure for a regression model is used with cross-section data.	16	K2	CO3
2.	Write a note on the following: (i) Test for Significance of Regression (ii) Tests on individual regression coefficients	16	K2	CO3
3.	Write a note on statistical procedure for estimation of confidence intervals on individual regression coefficients and the mean response.	16	K2	CO3
4.	Explain how a regression model can be used to predict future observations on the response variable.	16	K2	CO3
5.	Explain how residual plots and scaled residuals support in checking the adequacy of a model.	16	K2	CO3
6.	Explain in detail about the various techniques available for variable selection in regression.	16	K2	CO3
7.	Discuss about generalized and weighted least squares methods.	16	K2	CO3
8.	What is the use of Durbin Watson test and Cochrane – Orcutt method? Explain in detail.	16	K2	CO3